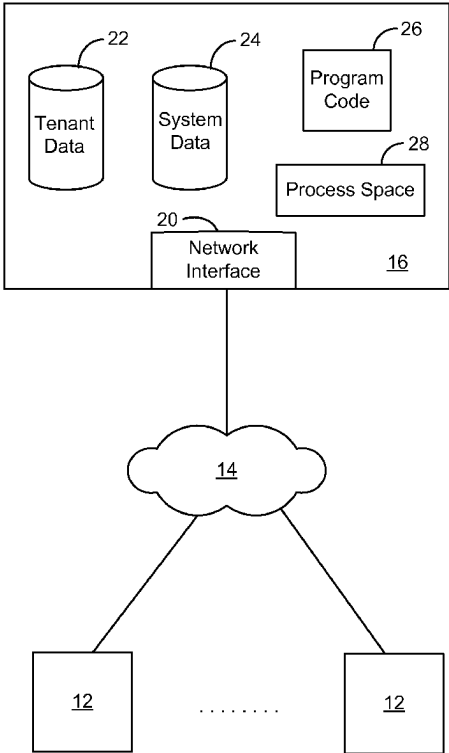


(54) INTEGRATING AND MANAGING SOCIAL NETWORKING INFORMATION IN AN ON-DEMAND DATABASE SYSTEM	Publication Classification
(71) Applicant: salesforce.com, inc. , San Francisco, CA (US)	(51) Int. Cl.
(72) Inventor: Chet Chauhan , San Francisco, CA (US)	<i>G06Q 30/0201</i> (2023.01)
(21) Appl. No.: 19/216,102	<i>G06F 16/95</i> (2019.01)
(22) Filed: May 22, 2025	<i>G06F 16/9535</i> (2019.01)
	<i>G06F 16/958</i> (2019.01)
	<i>G06Q 50/00</i> (2024.01)
	<i>H04L 51/216</i> (2022.01)
	<i>H04L 51/52</i> (2022.01)
	<i>H04L 67/10</i> (2022.01)
	(52) U.S. Cl.
	CPC <i>G06Q 30/0201</i> (2013.01); <i>G06F 16/95</i> (2019.01); <i>G06F 16/9535</i> (2019.01); <i>G06F 16/958</i> (2019.01); <i>G06Q 50/01</i> (2013.01); <i>H04L 51/216</i> (2022.05); <i>H04L 51/52</i> (2022.05); <i>H04L 67/10</i> (2013.01)

- Related U.S. Application Data**
- (63) Continuation of application No. 18/746,365, filed on Jun. 18, 2024, now Pat. No. 12,354,113, which is a continuation of application No. 18/331,563, filed on Jun. 8, 2023, now Pat. No. 12,045,834, which is a continuation of application No. 17/806,997, filed on Jun. 15, 2022, now Pat. No. 11,715,110, which is a continuation of application No. 17/249,956, filed on Mar. 19, 2021, now Pat. No. 11,386,434, which is a continuation of application No. 16/676,182, filed on Nov. 6, 2019, now Pat. No. 10,977,661, which is a continuation of application No. 15/374,807, filed on Dec. 9, 2016, now Pat. No. 10,489,790, which is a continuation of application No. 14/145,870, filed on Dec. 31, 2013, now Pat. No. 9,519,711, which is a continuation of application No. 12/917,326, filed on Nov. 1, 2010, now Pat. No. 8,621,015.
- (60) Provisional application No. 61/256,861, filed on Oct. 30, 2009.

(57) **ABSTRACT**

Some embodiments comprise integrating information from a social network into a multi-tenant database system. A plurality of information from the social network is retrieved, using a processor and a network interface of a server computer in the multi-tenant database system, wherein the plurality of information is associated with a message transmitted using the social network. Metadata related to the transmitted message is generated, using the processor. A conversation object is generated, using the processor, based on the plurality of information associated with the transmitted message and the metadata related to the transmitted message. The conversation object is then stored in an entity in the multi-tenant database system, using the processor of the server computer.



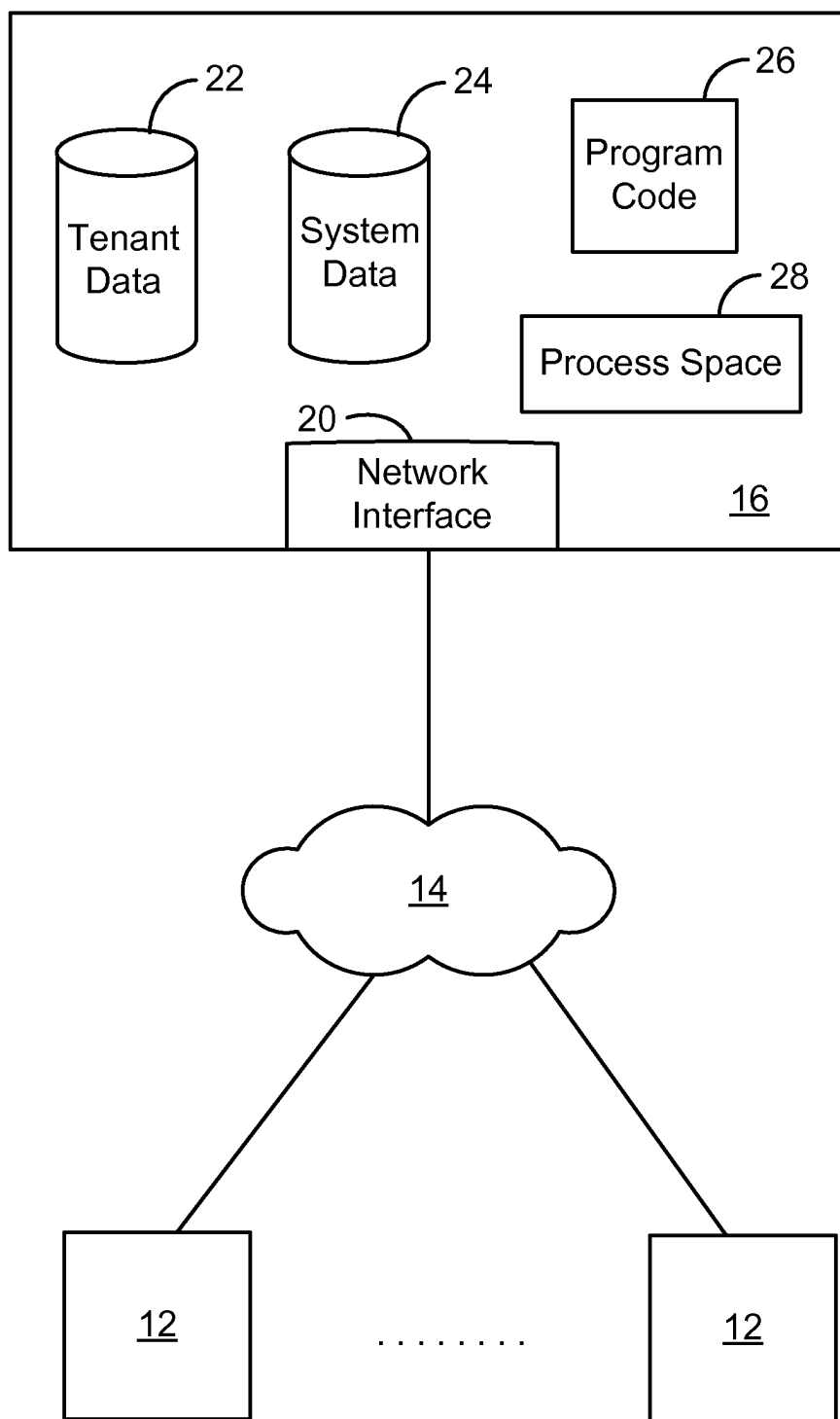


FIG. 1

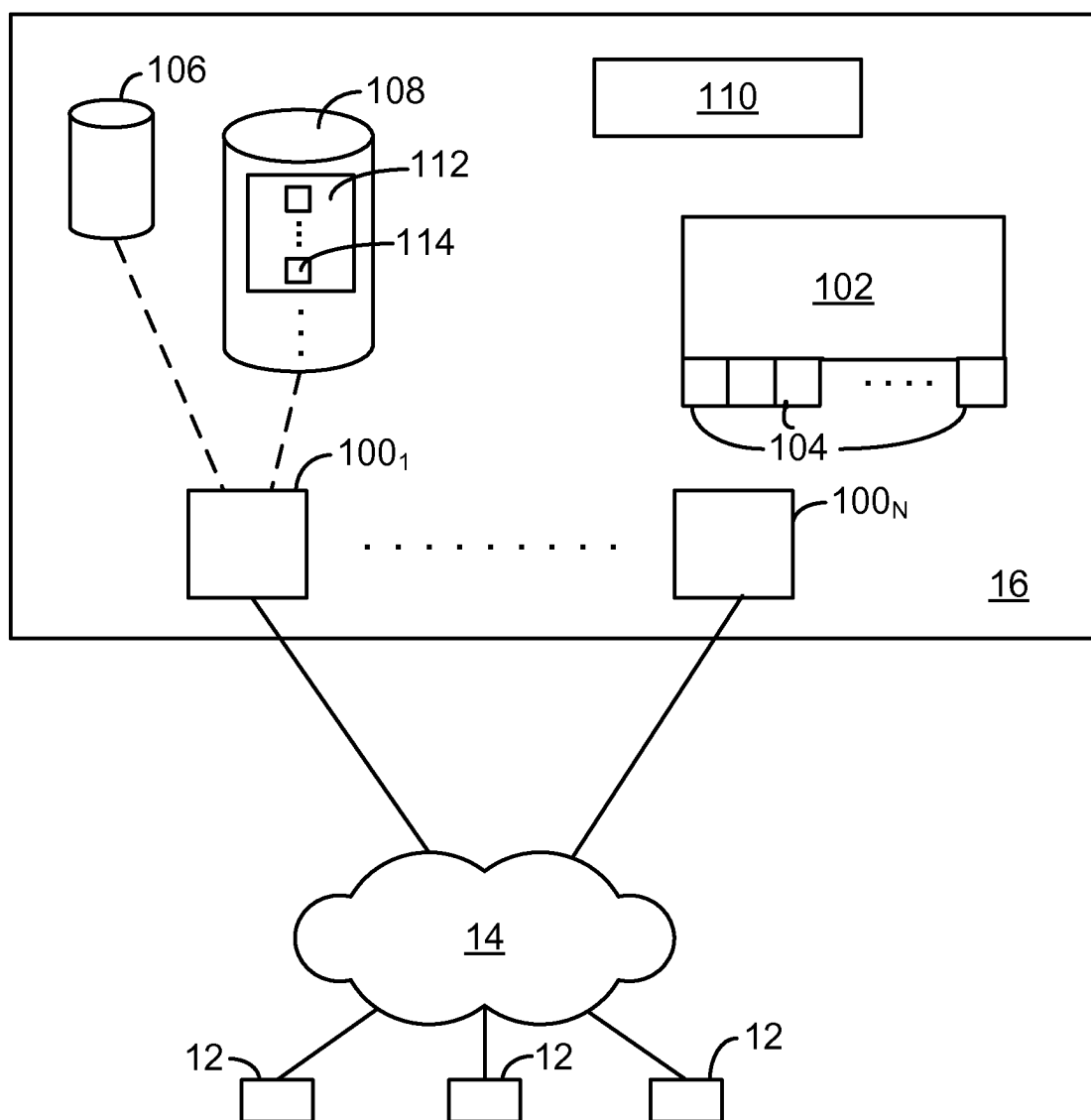


FIG. 2

200				
(.account) 202			data 203	
201	Org id	acc id	name	User ID Ownership
org #1	ood 1	acc 1	Enterprise Constellation	
	ood 1	acc 2		
		⋮		
org #2	ood 1			
	ood 2			
	ood 2			
⋮	⋮	⋮	⋮	⋮
org #N	oodN			
	oodN			

FIG. 3

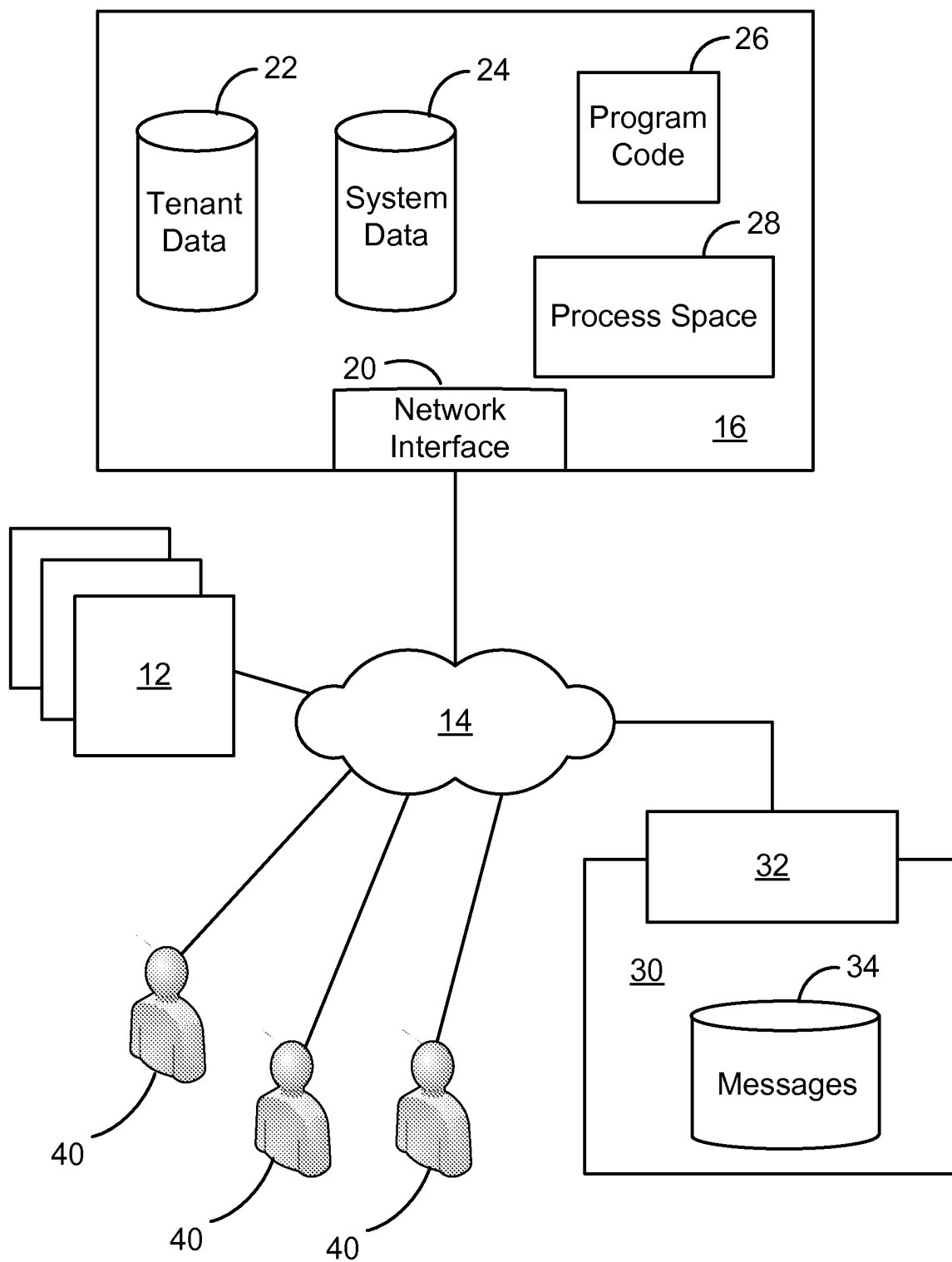


FIG. 4

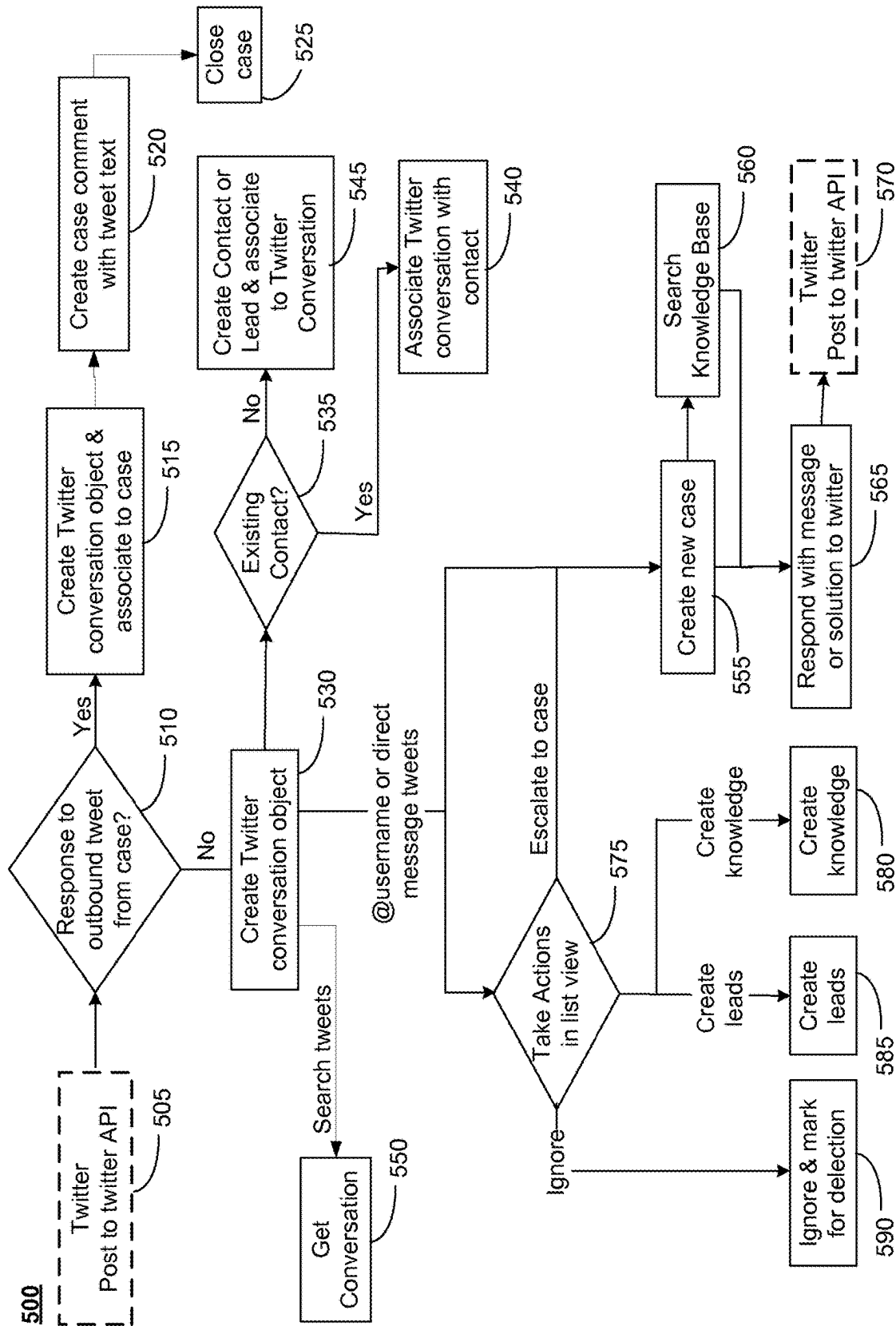


FIG. 5

INTEGRATING AND MANAGING SOCIAL NETWORKING INFORMATION IN AN ON-DEMAND DATABASE SYSTEM

INCORPORATION BY REFERENCE

[0001] An Application Data Sheet is filed concurrently with this specification as part of the present application. Each application that the present application claims benefit of or priority to as identified in the concurrently filed Application Data Sheet is incorporated by reference herein in its entirety and for all purposes.

BACKGROUND

[0002] Social networks, such as Twitter®, Facebook®, and Yammer® have grown in popularity not only for casual social purposes, but also for work-related purposes. Users of social networks post messages related to a company's products and/or services containing information that could be useful to understand customers, track market developments, manage public relations, and address issues related to the company's products and/or services, employees, or the company itself.

SUMMARY

[0003] The implementations disclosed herein relate generally to database systems and more particularly to integration of information from social networks into multi-tenant database systems ("MTS"). Embodiments of the implementations are particularly useful for integrating information from social networks such as twitter.com, yammer.com, and other similar services. The implementations described herein provide methods and systems for integrating information harvested from social networks (e.g., posted messages) into a MTS. Various embodiments provide methods and systems for enabling customer support representatives to manage social networks like any other support channel. Various embodiments provide methods and systems for enabling multiple agents to pull, filter, process, search through, and take action based on such harvested information. Various embodiments provide methods and systems for using templates for such harvested information and storing harvested information in a knowledge base to increase productivity.

[0004] Some embodiments comprise integrating information from a social network into a multi-tenant database system. A plurality of information from the social network is retrieved, using a processor and a network interface of a server computer in the multi-tenant database system, wherein the plurality of information is associated with a message transmitted using the social network. Metadata related to the transmitted message is generated, using the processor. A conversation object is generated, using the processor, based on the plurality of information associated with the transmitted message and the metadata related to the transmitted message. The conversation object is then stored in an entity in the multi-tenant database system, using the processor of the server computer.

[0005] Some embodiments comprise a non-transitory computer-readable medium tangibly encoded with instructions for performing the above-described operations and variations thereof.

[0006] Some embodiments comprise a system comprising a processor, a network interface, and a non-transitory com-

puter-readable medium tangibly encoded with instructions for performing the above-described operations and variations thereof.

[0007] Reference to the remaining portions of the specification, including the drawings and claims, will realize other features and advantages of the implementations. Further features and advantages of the implementations, as well as the structure and operation of various embodiments, are described in detail below with respect to the accompanying drawings. In the drawings, like reference numbers indicate identical or functionally similar elements.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic illustrating an environment including a MTS.

[0009] FIG. 2 is a schematic illustrating elements of a MTS and interconnections therein in an embodiment.

[0010] FIG. 3 is a schematic illustrating an example of objects represented as a main table in an embodiment.

[0011] FIG. 4 is a schematic illustrating an example system for integrating and managing social networking information in a MTS in an embodiment.

[0012] FIG. 5 illustrates an example process flow for an integration with the twitter social network.

DETAILED DESCRIPTION

[0013] The implementations described herein provide methods and systems for integrating information harvested from social networks (e.g., posted messages) into a MTS. Customer support representatives are able to manage social networks like any other support channel. Multiple agents can pull, filter, process, search through, and take action based on such harvested information. Templates can be used to characterize harvested information and store it into standard object types. Harvested information can be stored in a knowledge base to increase productivity.

Definitions

[0014] Each database can generally be viewed as a collection of objects, such as a set of logical tables, containing data fitted into predefined categories. A table, one representation of a data object, is used herein to simplify the conceptual description of objects and custom objects in the present disclosure. It should be understood that the terms table, object and entity may be used interchangeably herein. Each table generally contains one or more data categories logically arranged as columns or fields in a viewable schema. Each row or record of a table contains an instance of data for each category defined by the fields. For example, a CRM database may include a table that describes a customer with fields for basic contact information such as name, address, phone number, fax number, etc. Another table might describe a purchase order, including fields for information such as customer, product, sale price, date, etc.

[0015] As used herein, the term multi-tenant database system ("MTS") refers to those systems in which various elements of hardware and software of the database system may be shared between one or more customers. For example, a given application server may simultaneously process requests for a great number of customers, and a given database table may store rows for a potentially much greater number of customers. In some MTS embodiments, standard entity tables might be provided. For CRM database

applications, such standard entities might include tables for relevant data, such as Account, Contact, Lead and Opportunity, each containing predefined fields.

[0016] As used herein, the term social network or social networking site refers to any multiuser online system that enables its users to create and preserve connections, communicate by posting broadcast or private messages, and share information with other users of the online system. Examples of social networks include twitter, Facebook, and Yammer.

[0017] As used herein, the term message includes any message posted on a social network, including but not limited to status updates, private messages sent to one or more users of the social network, or public messages for broadcast to everyone or to a subgroup of users.

Multi-Tenant Database System (MTS) Overview

[0018] FIG. 1 illustrates an environment wherein a MTS might be used. As illustrated in FIG. 1 (and in more detail in FIG. 2) any user systems 12 might interact via a network 14 with a MTS 16. The users of those user systems 12 might be users in differing capacities and the capacity of a particular user system 12 might be entirely determined by the current user. For example, when a salesperson is using a particular user system 12 to interact with MTS 16, that user system has the capacities allotted to that salesperson. However, while an administrator is using that user system to interact with MTS 16, that user system has the capacities allotted to that administrator. Network 14 can be a local area network ("LAN"), wide area network ("WAN"), wireless network, point-to-point network, star network, token ring network, hub network, or other configuration. As the most common type of network in current use is a Transfer Control Protocol and Internet Protocol ("TCP/IP") network such as the global internetwork of networks often referred to as the "Internet" with a capital "I," that will be used in many of the examples herein, but it should be understood that the networks that the system might use are not so limited, although TCP/IP is the currently preferred protocol.

[0019] User systems 12 might communicate with MTS 16 using TCP/IP and, at a higher network level, use other common Internet protocols to communicate, such as Hypertext Transfer Protocol ("HTTP"), file transfer protocol ("FTP"), Andrew File System ("AFS"), wireless application protocol ("WAP"), etc. As an example, where HTTP is used, user system 12 might include a HTTP client commonly referred to as a "browser" for sending and receiving HTTP messages from a HTTP server at MTS 16. Such a HTTP server might be implemented as the sole network interface between MTS 16 and network 14, but other techniques might be used as well or instead. In some embodiments, the interface between MTS 16 and network 14 includes load-balancing functionality, such as round-robin HTTP request distributors to balance loads and distribute incoming HTTP requests evenly over a plurality of HTTP servers. Preferably, each of the plurality of HTTP servers has access to the MTS's data, at least as for the users that are accessing that server.

[0020] In aspects, the system shown in FIG. 1 implements a web-based customer relationship management ("CRM") system. For example, in one aspect, MTS 16 can include application servers configured to implement and execute CRM software applications as well as provide related data, program code, forms, web pages and other information to

and from user systems 12 and to store to, and retrieve from, a database system related data, objects and web page content. With a MTS, tenant data is preferably arranged so that data of one tenant is kept separate from that of other tenants so that one tenant does not have access to another's data, unless such data is expressly shared.

[0021] One arrangement for elements of MTS 16 is shown in FIG. 1, including a network interface 20, storage 22 for tenant data, storage 24 for system data accessible to MTS 16 and possibly multiple tenants, program code 26 for implementing various functions of MTS 16, and a process space 28 for executing MTS system processes and tenant-specific processes, such as running applications as part of an application service.

[0022] Some elements in the system shown in FIG. 1 include conventional, well-known elements that need not be explained in detail here. For example, each user system 12 could include a desktop personal computer, workstation, laptop, personal digital assistant ("PDA"), cell phone, or any WAP-enabled device or any other computing device capable of interfacing directly or indirectly to the Internet or other network connection. User system 12 typically runs a HTTP client, e.g., a browsing program, such as Microsoft's Internet Explorer® browser, Mozilla's Firefox® browser, Netscape's Navigator® browser, Apple's Safari® browser, the Opera® browser, or a WAP-enabled browser in the case of a cell phone, PDA, or other wireless device, or the like, allowing a user (e.g., subscriber of a CRM system) of user system 12 to access, process and view information and pages available to it from MTS 16 over network 14. Each user system 12 also typically includes one or more user interface devices, such as a keyboard, a mouse, touch screen, pen or the like, for interacting with a graphical user interface ("GUI") provided by the browser on a display (e.g., monitor screen, LCD display, etc.) in conjunction with pages, forms and other information provided by MTS 16 or other systems or servers. As discussed above, the system is suitable for use with the Internet, which refers to a specific global internetwork of networks. However, it should be understood that other networks can be used instead of the Internet, such as an intranet, an extranet, a virtual private network ("VPN"), a non-TCP/IP-based network, any LAN or WAN or the like.

[0023] According to one embodiment, each user system 12 and all of its components are operator configurable using applications, such as a browser, including program code run using a central processing unit such as an Intel Pentium® processor or the like. Similarly, MTS 16 (and additional instances of MTS's, where more than one is present) and all of their components might be operator configurable using application(s) including program code run using a central processing unit such as an Intel Pentium processor or the like, or multiple processor units. Program code for operating and configuring MTS 16 to intercommunicate and to process web pages and other data and media content as described herein is preferably downloaded and stored on a hard disk, but the entire program code, or portions thereof, may also be stored in any other volatile or non-volatile memory medium or device as is well known, such as a ROM or RAM, or provided on any media capable of storing program code, such as a compact disk ("CD") medium, digital versatile disk ("DVD") medium, a floppy disk, and the like. Additionally, the entire program code, or portions thereof, may be transmitted and downloaded from a software source, e.g., over the Internet, or from another server, as is well known,

or transmitted over any other conventional network connection as is well known (e.g., extranet, VPN, LAN, etc.) using any communication medium and protocols (e.g., TCP/IP, HTTP, HTTPS, WAP, Ethernet, etc.) as are well known. It will also be appreciated that program code for implementing aspects of the system can be implemented in any programming language that can be executed on a server or server system such as, for example, in C, C++, HTML, Java, JavaScript, WML, any other scripting language, such as VBScript and many other programming languages as are well known.

[0024] It should also be understood that each user system **12** may include differing elements, for example, one user system **12** might include a user's personal workstation running Microsoft's Internet Explorer browser while connected to MTS **16** by VPN, another user system **12** might include a thin-client netbook (e.g., Asus Eee PC®) running the Opera browser while connected to MTS **16** through an extranet, and another user system **12** might include a PDA running a WAP-enabled browser while connected to MTS **16** over third-party cellular networks.

[0025] According to one embodiment, each MTS **16** is configured to provide web pages, forms, data and media content to user systems **12** to support the access by user systems **12** as tenants of MTS **16**. As such, MTS **16** provides security mechanisms to keep each tenant's data separate unless the data is shared. If more than one MTS **16** is used, they may be located in close proximity to one another (e.g., in a server farm located in a single building or campus), or they may be distributed at locations remote from one another (e.g., one or more servers located in city A and one or more servers located in city B). Each MTS **16** may include one or more logically and/or physically connected servers distributed locally or across one or more geographic locations. Additionally, the term "server" is meant to include a computer system, including processing hardware and process space(s), and an associated storage system and database application (e.g., relational database management system ("RDBMS")), as is well known in the art. It should also be understood that "server system" and "server" are often used interchangeably herein. Similarly, the databases described herein can be implemented as single databases, a distributed database, a collection of distributed databases, a database with redundant online or offline backups or other redundancies, etc., and might include a distributed database or storage network and associated processing intelligence.

[0026] FIG. 2 illustrates elements of MTS **16** and various interconnections in an exemplary embodiment. In this example, the network interface is implemented as one or more HTTP application servers **100**. Also shown is system process space **102** including individual tenant process space (s) **104**, a system database **106**, tenant database(s) **108**, and a tenant management process space **110**. Tenant database **108** might be divided into individual tenant storage areas **112**, which can be either a physical arrangement or a logical arrangement. Within each tenant storage area **112**, a user storage **114** might similarly be allocated for each user.

[0027] It should also be understood that each application server **100** may be communicably coupled to database systems, e.g., system database **106** and tenant database(s) **108**, via a different network connection. For example, one application server **1001** might be coupled via the Internet **14**, another application server **100x-1** might be coupled via a direct network link, and another application server **100x**

might be coupled by yet a different network connection. TCP/IP is the currently preferred protocol for communicating between application servers **100** and the database system, however, it will be apparent to one skilled in the art that other transport protocols may be used to optimize the system depending on the network interconnect used.

[0028] In aspects, each application server **100** is configured to handle requests for any user/organization. Because it is desirable to be able to add and remove application servers from the server pool at any time for any reason, there is preferably no server affinity for a user and/or organization to a specific application server **100**. In one embodiment, therefore, an interface system (not shown) implementing a load-balancing function (e.g., an F5 Big-IP load balancer) is communicably coupled between the application servers **100** and the user systems **30** to distribute requests to the application servers **100**. In one aspect, the load balancer uses a least connections algorithm to route user requests to the application servers **100**. Other examples of load balancing algorithms, such as round robin and observed response time, also can be used. For example, in certain aspects, three consecutive requests from the same user could hit three different servers, and three requests from different users could hit the same server. In this manner, MTS **16** is multi-tenant, wherein MTS **16** handles storage of different objects and data across disparate users and organizations.

[0029] As an example of storage, one tenant might be a company that employs a sales force where each user (e.g., a salesperson) uses MTS **16** to manage their sales process. Thus, a user might maintain contact data, leads data, customer follow-up data, performance data, goals and progress data, etc., all applicable to that user's personal sales process (e.g., in tenant database **108**). In one MTS arrangement, since all of this data and the applications to access, view, modify, report, transmit, calculate, etc., can be maintained and accessed by a user system having nothing more than network access, the user can manage his or her sales efforts and cycles from any of many different user systems. For example, if a salesperson is visiting a customer and the customer has Internet access in their lobby, the salesperson can obtain critical updates as to that customer while waiting for the customer to arrive in the lobby.

[0030] While each user's sales data might be separate from other users' sales data regardless of the employers of each user, some data might be organization-wide data shared or accessible by a plurality of users or all of the sales force for a given organization that is a tenant. Thus, there might be some data structures managed by MTS **16** that are allocated at the tenant level while other data structures might be managed at the user level. Because an MTS might support multiple tenants including possible competitors, the MTS, in one implementation, has security protocols that keep data, applications, and application use separate. Also, because many tenants will opt for access to an MTS rather than maintain their own system, redundancy, up-time and backup are more critical functions and need to be implemented in the MTS.

[0031] In addition to user-specific data and tenant-specific data, MTS **16** might also maintain system-level data usable by multiple tenants or other data. Such system-level data might include industry reports, news, postings, and the like that are sharable among tenants.

[0032] In certain aspects, user systems **30** communicate with application servers **100** to request and update system-

level and tenant-level data from MTS 16; this may require one or more queries to system database 106 and/or tenant database 108. MTS 16 (e.g., an application server 100 in MTS 16) automatically generates one or more SQL statements (a SQL query) designed to access the desired information.

[0033] FIG. 3 illustrates an example of an object represented as a main table 200 that holds data items for multiple tenants. In the particular example shown in FIG. 3, the main table 200 (.account) represents a standard Account entity that holds account information for multiple organizations. As shown, main table 200 includes an organization ID column 201 and an account ID column 202 that acts as the primary key for main table 200. For example, for organization #1, column 201 shows that Org ID="ood 1," and two account IDs are listed ("ood 1" and "ood 2"). Main table 200 also includes a plurality of data columns 203 containing other information about each row. For example, for account ID "ood1," information is stored about the account name (i.e., "Enterprise"). Main table 200 may also include column 209 that stores the user ID of the user that owns or created the specific account that is stored in that row.

[0034] The organization ID column 201 is provided to distinguish among organizations using the MTS. As shown, N different organizations have data stored in main table 200. In an exemplary embodiment, the organization IDs in column 201 are defined as Char (15), but may be defined as other data types. In one embodiment, the first 3 characters of the organization ID is set to a predefined prefix, such as "ood", although another subset of characters in the organization ID may be used to hold such a prefix if desired.

[0035] In the particular example of FIG. 3, where the table represents a standard entity, data columns 203 are predefined data columns, or standard fields, that are provided to the various organizations that might use the table. In the Account entity example described above, such standard fields might include a name column, a site column, a number of employees column and others as would be useful for storing account-related information. Each of the data columns 203 is preferably defined to store a single data type per column.

Integration Environment

[0036] FIG. 4 illustrates a MTS environment for integrating and managing information pulled from a social network, such as twitter, Facebook, or Yammer. As illustrated in FIG. 4, social network 30 is accessible by MTS 16 via a network 14. The members 40 of social network 30 also access social network 30 via network 14. Social network 30 is able to receive a message from a first member 40 that is then transmitted to one or more members 40 of social network 30. Messages may be stored in a data store 34. An interface 32 may be provided in order to provide a user interface for members 40. Interface 32 may also provide an API through which MTS 16 is able to retrieve the messages in data store 34 and transmit both new messages and message replies through the social network 30. Members 40 may be able to access social network 30 using any conventional Internet-connected device, including but not limited to a personal computer, a cellphone, a portable computing device, etc.

[0037] In an example embodiment, social network 30 is the twitter social network. A member 40 of twitter can broadcast a message (i.e., tweet) to all members of the twitter social network who are following tweets by the

member 40. A member 40 of twitter can also send a targeted message to a single, identified member of twitter, or to a group of members.

[0038] In an example embodiment, the MTS periodically polls the social network for new transmissions of messages and pulls information related to each newly transmitted message; this information is stored in the MTS in the form of a Conversation object. In an example embodiment wherein social network 30 is twitter, the MTS polls twitter's search API on a regular basis and stores all search results as twitter Conversation objects. Each twitter Conversation object is stored as a record in an entity of the MTS. An exemplary embodiment of a Conversation object for twitter is described in the table below.

Example Embodiment of a Conversation Object for Twitter

Field	Data Type	Description
Twitter Conversation Id	Auto Number	The Row Id where the Conversation object is stored in an entity of the MTS.
Twitter Id	Text(100) (External Id)	The Id of the message on twitter-pulled back via the API.
Content	Text[140]	The actual contents of the message (e.g., RT @EBF2010 European Banking Forum event looking at how to drive revenue; customer solutions, operational efficiency, and risk management). For twitter, this field may be limited to 140 characters.
Published Date	Date Time	The date and time the message was originally tweeted on twitter.
Author Name	Text(255)	The text name of the person that sent the message.
Twitter Profile	URL(255)	A URL to the author's twitter profile.
Type	Picklist	The source of the message, e.g., @reply, direct message, and Search.
Status	Picklist	The status or disposition of the conversation. As users process each conversation, they will disposition the conversation as to what happened. Values include: no action, ignore, case created, promoted to knowledge, solved by community, promoted to lead, etc.

-continued

Field	Data Type	Description
Avatar Icon	Text field with URL to the avatar image	Shows the user's image on twitter. The icon may be resized to a fixed standard size so that it fits nicely in a List View or related list. Typically this field will be displayed next to the user's name.
Case Id	Lookup	This is a foreign key to the Case that is filled in for any twitter conversation that is promoted to a Case.
Contact Id	Lookup	This is a foreign key to the Contact object. If the message is linked to an existing contact with a twitter username, this field should be set to the contact the message was associated with.
Lead Id	Lookup	If the message is "promoted" into a lead, this should specify the field LeadId that the message was converted into.
Solution Id	Lookup	A foreign key for the solution for any message that is promoted to a solution.
Twitter Search Setup Id	Lookup	A foreign key to the twitter search setup record that created this twitter Conversation Object
Owner	Lookup to User or Group	The person(s) or queue that is assigned to look at this twitter conversation; this may be specified in the twitter search setup where you specify the owner for the results.
Campaign Id	Lookup	Link to Campaign record
In Reply to Status Id	Text(255)	Hidden field used to keep track of twitter threads
In Reply to Use Id	Text(255)	Hidden field used to keep track of twitter threads
Solution Created	Hyperlink formula field	Icon to show that a solution is attached to this twitter conversation
Case Created	Hyperlink formula field	Icon to show that a case is attached to this twitter conversation
Lead Created	Hyperlink formula field	Icon to show that a lead is attached to this twitter conversation

Integrating Information From A Social Network Into A MTS

[0039] FIG. 5 illustrates an example process flow **500** for one embodiment of an integration with the twitter social network. A message (i.e., tweet) is transmitted via the twitter social network, so when the MTS polls twitter's search API, the newly transmitted message is detected (**505**). In one embodiment, a user can define message filters to refine the search results.

[0040] If the message is a response to an earlier outbound message related to an existing case (**510**), then the user may create a Conversation object in the MTS and associate it with the existing case (**515**). The user can then create a case comment using the message text (**520**), and the case can be closed if it is appropriate to do so (**525**).

[0041] If the message is not a response to an earlier outbound message related to an existing case (**510**), then the user may create a new Conversation object (**530**) in the MTS. The MTS may then search through existing contacts and/or leads to see if the message was sent by an existing contact (**535**). If so, the MTS may associate the Conversation object with the existing contact (**540**). In one embodiment, if more than one existing contact and/or lead appears in the search results, then one of the following actions may be taken: (1) no association will be created, or (2) associations are created with all existing contacts and/or leads appearing in the search results, or (3) an association will be created with the most recent contact or lead. If the message was not sent by an existing contact (**535**), the MTS may create a new Contact or Lead and associate it with the Conversation object (**545**). In one embodiment, the Contact or Lead object includes a custom field to store the twitter username for the new contact.

[0042] If the message is a search tweet (i.e., a publicly broadcast status update), the MTS may retrieve the prior conversation, if any (**550**). If the message is an @username tweet (i.e., a publicly accessible targeted message) or direct message (i.e., a private message), the user may be able to determine that the content of the message warrants creation of a new case and create a new case (**555**) and search the knowledge base (**560**) to see if there is an existing solution or any further information. If there is, then the user may be able to respond to the message to communicate the solution or further information (**565**)—the MTS may post the response to the twitter API (**570**).

[0043] In one embodiment, the user may be able to view a plurality of messages in a list view, select particular messages, and take actions upon the selected messages in the list (**575**). The user may be able to escalate a selected message to a case and create a new case (**555**). The user may be able to use the selected message to create a new knowledge base article (**580**). The user may be able to use the selected message to create a new lead (**585**). Or, the user may be able to choose to ignore the selected message and mark it for deletion (**590**). In one embodiment, the user may be able to change the status of the selected message (e.g., "solved by community") or change other metadata related to the selected message.

[0044] In one embodiment, when a new Conversation object is created based on a transmitted message, metadata related to the transmitted message is generated and stored along with the Conversation object in the MTS. In one embodiment, when a new Conversation object is created, it can be categorized in accordance with any of a number of aspects, including but not limited to the message content, the

author of the message, the recipient of the message, the date and/or time of the message, the case or lead or knowledge base article with which the message is associated, etc.

[0045] In one embodiment, messages from certain users (e.g., usernames belonging to MTS system administrators) may be filtered out in the search results.

User Interface

[0046] In one embodiment, the search for newly transmitted messages may be triggered manually. In one embodiment, the search for newly transmitted messages may be run automatically, possibly on a periodic basis. In one embodiment, the user can enable auto-creation of cases, leads, and/or contact

[0047] In one embodiment, outbound messages destined for the social network can be generated from several locations in the user interface-when the user is viewing a case, a Conversation object, a contact, and/or a lead. In one embodiment, the user interface allows creation of an outbound message from within the MTS user interface. In one embodiment, the user interface allows creation of an outbound message directly from the social network's user interface.

[0048] In one embodiment, once an action has been taken on a message (e.g., create case, create lead, ignore), the user interface indicates that an action has been taken on the message; in one embodiment, the user interface indicates the exact disposition of the message.

[0049] In one embodiment, reports can be generated to determine various statistics, which may include: how many messages were pulled in by the search, how many messages were reviewed, how many messages were promoted to cases or leads or knowledge base articles, how many messages were ignored, and how many messages are left to be dealt with.

Setup and Administration

[0050] In one embodiment, a Search object is created in order to define various search terms for use when searching for messages in the social network. In one embodiment, multiple instances of Search objects may be created. Each search instance may have an associated owner (either username or queue). All conversations that are pulled in as a result of the Search object may be assigned to the owner associated with the search object. For example, conversations pulled in using the search terms "comcast issue"+ "comcast problem" are assigned to the owner "twitter service." In another example, conversations pulled in using the search terms "Verizon offer"+ "which cable company" are assigned to the owner "twitter sales."

Example Embodiment of a Search Object for Twitter Integration

Twitter Search TermId	AutoNumber	The Row Id for the twitter search in salesforce.
Twitter Keyword Search	Text Area(255)	The Twitter search term
Owner	Lookup	Lookup to User/Queue. All incoming twitter conversations from this search will be assigned to this queue.

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Unknown twitter usernames	Picklist [Do nothing, Create Contact, Create Lead]	If "Create Contact", Contacts will automatically be created for unknown twitter users associated with Conversations. If "Create Lead", Leads will automatically be created for unknown twitter users associated with Twitter Conversations. If Do nothing", no object is created for unknown twitter users associated with Twitter Conversations. If the twitter user exists as a known contact then it should not create a contact or lead Options:
Lead Association Setting	Picklist	
Twitter Max Twitter ID	Text(255)	Hidden field
Twitter Max Message ID	Text(255)	Hidden field
Twitter Max Search Id	Text(255)	Hidden field
Twitter Account ID	Text(255)	Hidden field
Last Run Status	Picklist	Hidden field
Error Msg	TextArea(255)	Hidden field
Next API Reset Time	Date/Time	Hidden field
Twitter Max In Progress Id	Text(255)	Hidden field

[0051] In one embodiment, an Account object is created in order to enable system administrators to keep track of necessary user account information, wherein the user account is related to the social network.

Example Embodiment of an Account Object for Twitter Integration

Twitter Account Id	AutoNumber	The Row Id for the twitter account in salesforce.
Username	Text (100)	twitter username
Password	Text(50)	twitter password
Owner	Lookup	User/Queue lookup: this is the owner of the twitter account. It can be a user or a queue. If private, then this can be used to control visibility to the account.
Enable Auto-Case Creation	Checkbox	When checked, Cases and Contacts will automatically be created for incoming twitter replies and direct messages.
Unknown twitter usernames	Picklist [Do nothing, Create Contact, Create Lead]	If "Create Contact", Contacts will automatically be created for unknown twitter users associated with, twitter conversations. If "Create Lead", Leads will automatically be created for unknown twitter users associated with Twitter Conversations. If "Do nothing", no object is created for unknown twitter users associated with Twitter Conversations.

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Bit.ly username	Text(50)	If the twitter user exists as a known contact then it should not create a contact or lead
Bit.ly password	Text(100)	Username needed to create the shortened URL Stores the apiKEY to access the bit.ly API and generate the shortened URL.

[0052] While the invention has been described by way of example and in terms of the specific embodiments, it is to be understood that the invention is not limited to the disclosed embodiments. To the contrary, it is intended to cover various modifications and similar arrangements as would be apparent to those skilled in the art. Therefore, the scope of the appended claims should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements.

1-20. (canceled)

21. A system comprising:

a server system comprising one or more processors, the server system configurable to cause:

displaying, in a user interface (UI) on a display device, data of an account record stored in a customer relationship management (CRM) database system;
identifying one or more social media posts displayable in the UI;

communicating, responsive to the identifying, the one or more social media posts to the CRM database system;

assigning, in the CRM database system, one or more social media users associated with the one or more social media posts, including mapping the one or more social media users to a CRM record in the CRM database system;

providing an interface element in the UI for interacting with the one or more social media users; and
tracking, using the CRM record in the CRM database system, interactions from the interface element.

22. The system of claim **21**, wherein the interactions include a response to a post in an external social media network.

23. The system of claim **21**, wherein the CRM record includes data from the response.

24. The system of claim **21**, wherein the interface element is provided at least in part by the CRM database system.

25. The system of claim **21**, wherein the account record is the CRM record.

26. The system of claim **21**, wherein the mapping of the one or more social media users to the CRM record in the CRM database system includes matching the one or more social media users to a CRM contact stored in the CRM database system.

27. The system of claim **21**, the server system further configurable to cause:

triggering one or more follow-up actions, including creating a service request in the CRM database system, the one or more follow-up actions being trackable using the CRM database system.

28. A non-transitory computer-readable medium storing computer program instructions capable of being executed by one or more processors, the instructions configurable to cause:

displaying, in a user interface (UI) on a display device, data of an account record stored in a customer relationship management (CRM) database system;
identifying one or more social media posts displayable in the UI;

communicating, responsive to the identifying, the one or more social media posts to the CRM database system;
assigning, in the CRM database system, one or more social media users associated with the one or more social media posts, including mapping the one or more social media users to a CRM record in the CRM database system;

providing an interface element in the UI for interacting with the one or more social media users; and
tracking, using the CRM record in the CRM database system, interactions from the interface element.

29. The non-transitory computer-readable medium of claim **28**, wherein the interactions include a response to a post in an external social media network.

30. The non-transitory computer-readable medium of claim **29**, wherein the CRM record includes data from the response.

31. The non-transitory computer-readable medium of claim **28**, wherein the interface element is provided at least in part by the CRM database system.

32. The non-transitory computer-readable medium of claim **28**, wherein the account record is the CRM record.

33. The non-transitory computer-readable medium of claim **28**, wherein the mapping of the one or more social media users to the CRM record in the CRM database system includes matching the one or more social media users to a CRM contact stored in the CRM database system.

34. The non-transitory computer-readable medium of claim **28**, the instructions further configurable to cause:

triggering one or more follow-up actions, including creating a service request in the CRM database system, the one or more follow-up actions being trackable using the CRM database system.

35. A computer-implemented method comprising:
displaying, in a user interface (UI) on a display device, data of an account record stored in a customer relationship management (CRM) database system;
identifying one or more social media posts displayable in the UI;

communicating, responsive to the identifying, the one or more social media posts to the CRM database system;
assigning, in the CRM database system, one or more social media users associated with the one or more social media posts, including mapping the one or more social media users to a CRM record in the CRM database system;

providing an interface element in the UI for interacting with the one or more social media users; and
tracking, using the CRM record in the CRM database system, interactions from the interface element.

36. The computer-implemented method of claim **35**, wherein the interactions include a response to a post in an external social media network.

37. The computer-implemented method of claim **36**, wherein the CRM record includes data from the response.

38. The computer-implemented method of claim **35**, wherein the interface element is provided at least in part by the CRM database system.

39. The computer-implemented method of claim **35**, wherein the account record is the CRM record.

40. The computer-implemented method of claim **35**, wherein the mapping of the one or more social media users to the CRM record in the CRM database system includes matching the one or more social media users to a CRM contact stored in the CRM database system.

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